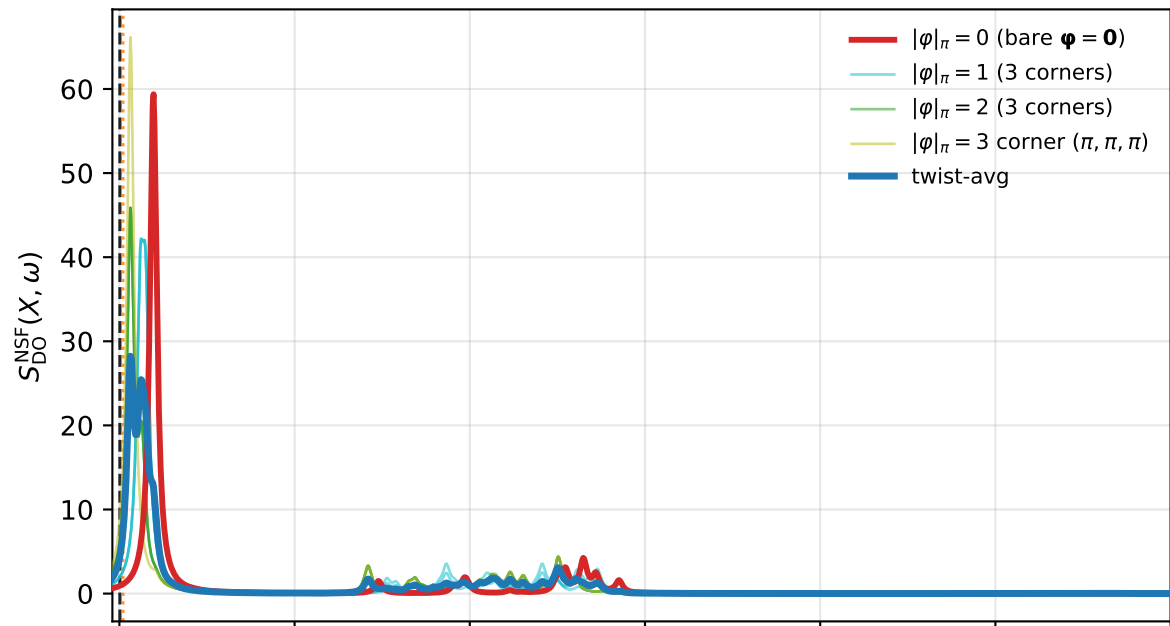
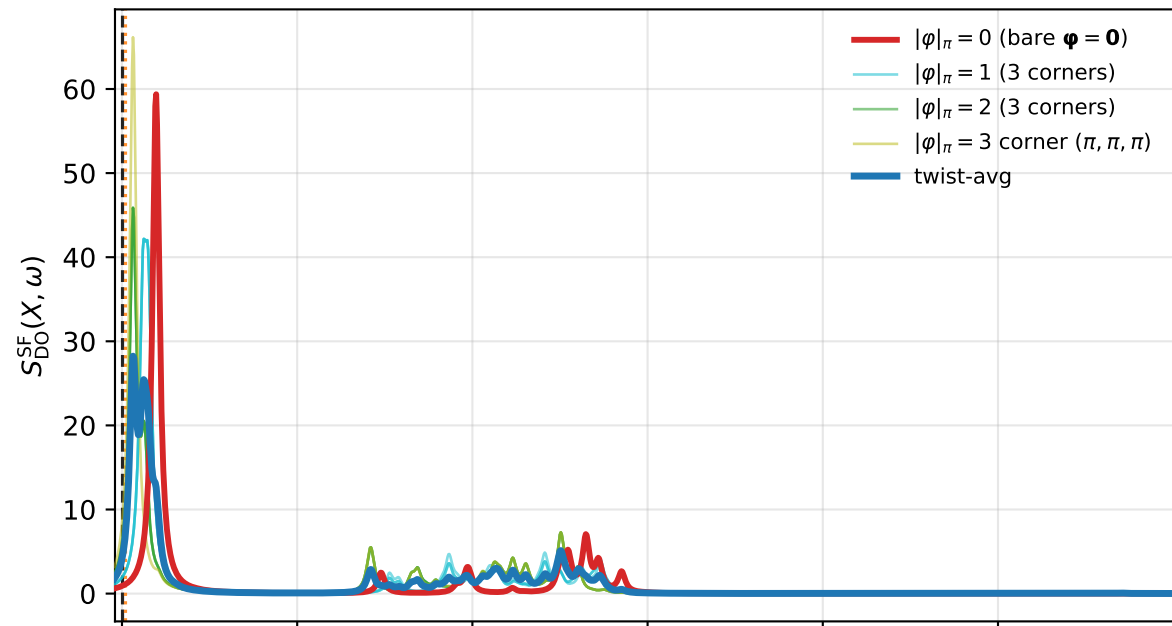


Per-corner DO-QSI NSF/SF on the 16-site cluster ($J_{\pm} = +0.05J_{zz}$): each $\boldsymbol{\varphi}$ corner coloured by parity class $|\varphi|_{\pi}$, twist-averaged in heavy blue.

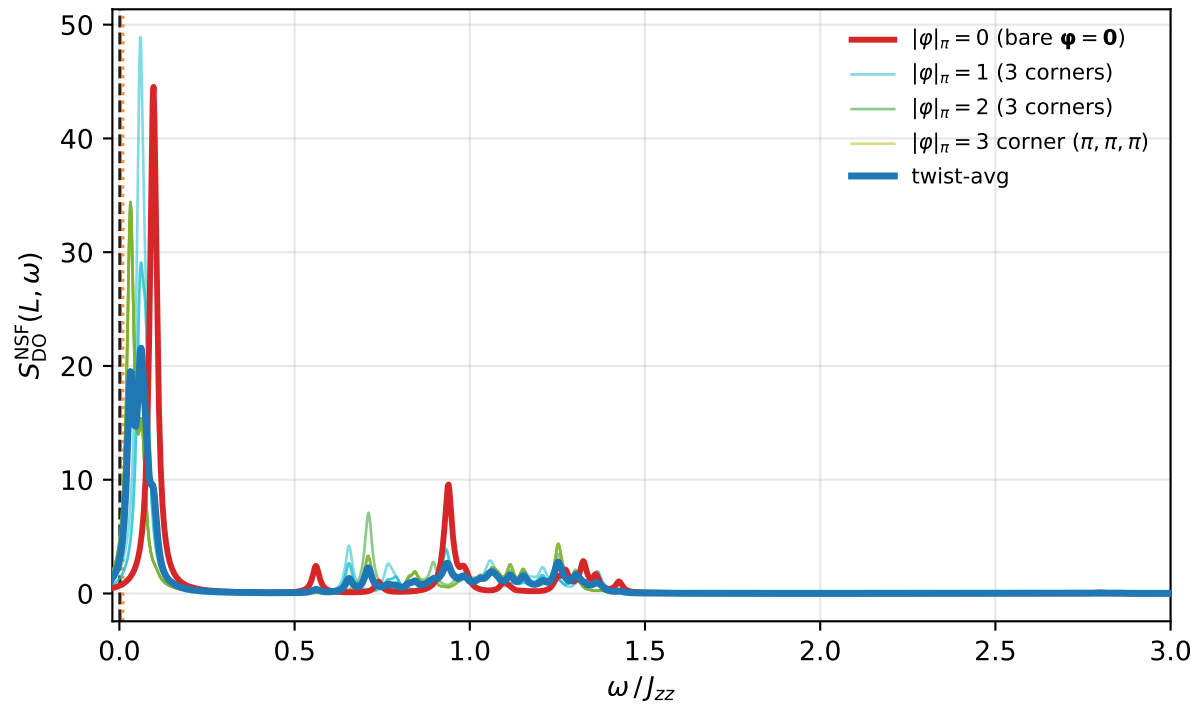
(a) NSF, $\mathbf{q} = X$, per corner



(b) SF, $\mathbf{q} = X$, per corner



(c) NSF, $\mathbf{q} = L$, per corner



(d) SF, $\mathbf{q} = L$, per corner

